Curvature from Non-Uniform Event-Order Geometry

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Abstract

Papers XXIII–XXV derived event ordering, observer-dependent kinematics, and separation geometry from phase-overlap structure without assuming spacetime. The present paper addresses the next question: when is that geometry uniform, and when is it distorted?

Flatness is defined as local uniformity of event neighborhood profiles in the transition graph: a region is flat if nearby events have identical local accessibility profiles. Curvature is defined as deviation from this local uniformity. Three curvature candidates are constructed and compared: (A) volume-growth distortion — deviation of metric-ball sizes from a local reference; (B) branching asymmetry — non-uniformity of local connectivity in the transition graph; (C) geodesic concentration — preferential routing of shortest paths through specific events.

Two toy models are worked explicitly: a symmetric two-layer model (flat, all curvature indicators vanish) and an asymmetric three-layer model with a bottleneck (curved, all three indicators are nonzero). In the curved model, geodesics concentrate through the bottleneck, providing a free-fall analogue: motion follows the accessibility structure of the event geometry without any imposed force.

The paper does not derive Einstein equations or curvature tensors. It establishes a concrete curvature precursor on the event-order framework, showing that non-uniform event accessibility produces detectable geometric distortion.

1 Introduction

Paper XXV [Smart, 2026] established three separation constructions on the event poset: chain-length separation for comparable events, observer-disagreement for incomparable events, and a transition-cost metric on the full Hasse graph. These give the event set a metric-like geometry.

The next question is: is that geometry uniform everywhere, or can it be distorted? In smooth general relativity, the answer is curvature: the Riemann tensor measures how the geometry deviates from flatness. The present paper constructs the discrete analogue: curvature as non-uniformity of local event accessibility.

Central principle

In VFD, curvature is the failure of local event accessibility to remain uniform.

Scope

This paper defines flatness, constructs curvature indicators, and demonstrates them on toy models. It does not derive Einstein equations, curvature tensors, or continuous geometry. It establishes a curvature precursor on the discrete event-order framework.

2 Local Neighborhoods and Flatness

2.1 Metric balls

Definition 1 (Metric ball)**.** *For event $e \in \mathcal{E}$ and radius $r \geq 0$, the metric ball in the transition graph $\mathcal{G}$ is:*

$$B_r(e) = \{e' \in \mathcal{E} : d_{\text{trans}}(e, e') \leq r\} \tag{1}$$

The local volume is $V_e(r) = |B_r(e)|$.

2.2 Local neighborhood profile

Definition 2 (Neighborhood profile)**.** *The neighborhood profile of event e is the sequence:*

$$\mathbf{V}(e) = (V_e(0), V_e(1), V_e(2), \ldots) \tag{2}$$

2.3 Flatness

Definition 3 (Local flatness)**.** *A subset $S \subset \mathcal{E}$ is locally flat at radius r if all events in S have identical neighborhood profiles up to radius r:*

$$V_e(k) = V_{e'}(k) \quad \text{for all } e, e' \in S \text{ and all } k \leq r \tag{3}$$

The event set $\mathcal{E}$ is globally flat if it is locally flat at radius r for all $r \leq \operatorname{diam}(\mathcal{G})/2$.

Remark 4. *Local flatness means that the transition graph has uniform volume-growth profiles from every event in S out to radius r. This is a discrete analogue of local homogeneity in Riemannian geometry (though it is weaker than local graph isomorphism).*

3 Curvature Candidate A: Volume-Growth Distortion

Definition 5 (Volume-growth curvature)**.** *For an event e at radius r, with a reference volume profile $V_{\text{ref}}(r)$ (from a designated flat/symmetric region or the event-set average), define:*

$$K_{\text{grow}}(e; r) = \frac{V_{\text{ref}}(r) - V_e(r)}{V_{\text{ref}}(r)} \tag{4}$$

Remark 6. *$K_{\text{grow}} = 0$: event e has the same local volume as the reference (flat). $K_{\text{grow}} > 0$ indicates slower-than-reference local growth, while $K_{\text{grow}} < 0$ indicates faster-than-reference local growth. These signs are heuristically analogous to positive and negative curvature, respectively, but no smooth-curvature identification is claimed here.*

4 Curvature Candidate B: Branching Asymmetry

Remark 7. *The directed Hasse graph (with edges $e_1 \to e_2$ when $e_1 \lessdot e_2$) also carries in-degree and out-degree information. The undirected definition above is preferred for the branching curvature indicator because it ensures the flat reference model has genuinely vanishing curvature. Directed branching asymmetry (in-degree vs out-degree) may serve as an additional causal-anisotropy indicator in future work.*

Definition 8 (Branching curvature)**.**

$$K_{\text{branch}}(e) = \left|\deg_{\mathcal{G}}(e) - \overline{\deg}\right| \tag{5}$$

where $\deg_{\mathcal{G}}(e)$ is the degree of e in the undirected transition graph and $\overline{\deg}$ is the mean degree over $\mathcal{E}$.

Remark 9. $K_{\text{branch}} = 0$*: the event has exactly average connectivity (locally regular).* $K_{\text{branch}} > 0$*: the event has anomalous connectivity — either a bottleneck (much higher degree than average) or a peripheral event (much lower degree). This measures structural non-uniformity in event accessibility.*

5 Curvature Candidate C: Geodesic Concentration

Definition 10 (Geodesic betweenness)**.** *For event e, define the geodesic betweenness centrality:*

$$\beta(e) = \sum_{\substack{s,t \in \mathcal{E} \\ s \neq e \neq t}} \frac{\sigma_{st}(e)}{\sigma_{st}} \tag{6}$$

where σ_{st} is the number of shortest paths from s to t in $\mathcal{G}$, and $\sigma_{st}(e)$ is the number that pass through e.

Definition 11 (Geodesic-concentration curvature)**.**

$$K_{\text{geo}}(e) = \frac{\beta(e) - \bar{\beta}}{\bar{\beta}} \tag{7}$$

where $\bar{\beta}$ is the mean betweenness over $\mathcal{E}$.

Remark 12. $K_{\text{geo}} = 0$*: the event carries its fair share of geodesics (uniform routing).* $K_{\text{geo}} > 0$*: the event is a bottleneck — geodesics concentrate through it. This is the free-fall analogue: paths through the event structure are drawn toward high-betweenness regions, just as geodesics in GR are drawn toward massive bodies.*

6 Geodesics and Free Fall

Definition 13 (Geodesic)**.** *A geodesic between events e_1 and e_2 is a shortest path in $(\mathcal{E}, d_{\text{trans}})$.*

Remark 14. *In a flat event geometry (all curvature indicators zero), geodesics are uniformly distributed: no event carries a disproportionate share of shortest paths. In a curved*

event geometry (nonzero curvature indicators), geodesics concentrate through specific regions.

This concentration is the structural analogue of gravitational free fall: objects (worldlines, chains) in the event structure are drawn toward regions of high accessibility, not by an imposed force, but by the geometry of the event graph itself. The "force" of gravity, in this picture, is the gradient of event accessibility.

7 Flat Toy Model: Symmetric Two-Layer

Example 15 (Symmetric pentagon model)**.** *The dual-pentagon model from Papers XXIII–XXV: 5 events in Layer 0, 5 in Layer 1, with every Layer-0 event covering every Layer-1 event. The Hasse graph is $K_{5,5}$ (complete bipartite).*

Neighborhood profiles. *Every Layer-0 event e has: $V_e(0) = 1$, $V_e(1) = 6$ (itself + 5 Layer-1 events), $V_e(2) = 10$ (all events). By symmetry, every Layer-1 event has the same profile.*

Curvature indicators.

- $K_{\text{grow}}(e; 1) = 0$ *for all e (all events have the same ball size).*
- $K_{\text{branch}}(e) = 0$ *for all e (every vertex in $K_{5,5}$ has undirected degree 5; $\overline{\deg} = 5$).*
- $K_{\text{geo}}(e) = 0$ *for all e (by the vertex-transitive symmetry of $K_{5,5}$ with equal partition sizes, all vertices have equal betweenness).*

Conclusion. *The symmetric two-layer model is flat: all curvature indicators vanish. This establishes the baseline.*

8 Curved Toy Model: Asymmetric Three-Layer with Bottleneck

Example 16 (Bottleneck model)**.** *Introduce a three-layer event graph:*

- ***Layer 0****: 4 events $\{a_1, a_2, a_3, a_4\}$.*
- ***Layer 1****: 1 event $\{b\}$ (the bottleneck).*
- ***Layer 2****: 4 events $\{c_1, c_2, c_3, c_4\}$.*

Cover relations: *$a_i \lessdot b$ for all i, and $b \lessdot c_j$ for all j. No other cover relations.*

Hasse graph: *A "bowtie" or hourglass — all Layer-0 events connect to the single bottleneck, which connects to all Layer-2 events.*

Partial order: *$a_i \prec b \prec c_j$ for all i, j. Within each layer, events are incomparable.*

Neighborhood profiles

- *a_i: $V_{a_i}(0) = 1$, $V_{a_i}(1) = 2$ (itself + b), $V_{a_i}(2) = 6$ (itself + b + 4 c-events).*
- *b: $V_b(0) = 1$, $V_b(1) = 9$ (itself + 4 a's + 4 c's), $V_b(2) = 9$ (all events).*
- *c_j: $V_{c_j}(0) = 1$, $V_{c_j}(1) = 2$ (itself + b), $V_{c_j}(2) = 6$.*

The profiles differ: $V_b(1) = 9 \neq V_{a_i}(1) = 2$. ***Not flat.***

Curvature indicators

Volume-growth curvature *(with* $V_{\text{ref}}(1) = \bar{V}(1) = (4 \times 2 + 1 \times 9 + 4 \times 2)/9 = 25/9 \approx 2.78$*):*

- $K_{\text{grow}}(a_i; 1) = (2.78 - 2)/2.78 = 0.28 > 0$ *(smaller than average — positive curvature).*
- $K_{\text{grow}}(b; 1) = (2.78 - 9)/2.78 = -2.24 < 0$ *(larger than average — negative curvature).*
- $K_{\text{grow}}(c_j; 1) = 0.28 > 0$.

Branching curvature. *Undirected degrees:* $\deg(a_i) = 1$, $\deg(b) = 8$, $\deg(c_j) = 1$. *Mean degree:* $\overline{\deg} = (4 \times 1 + 1 \times 8 + 4 \times 1)/9 = 16/9 \approx 1.78$.

- $K_{\text{branch}}(a_i) = |1 - 16/9| = 7/9 \approx 0.78$.
- $K_{\text{branch}}(b) = |8 - 16/9| = 56/9 \approx 6.2$ *(highest).*
- $K_{\text{branch}}(c_j) = |1 - 16/9| = 7/9 \approx 0.78$.

The bottleneck b *has by far the highest branching curvature.*

Geodesic concentration. *Every shortest path between a Layer-0 event and a Layer-2 event passes through* b*. There are* $4 \times 4 = 16$ *such pairs, and* b *lies on all 16 geodesics. Paths between same-layer events also pass through* b *(e.g.* $a_1 \to b \to a_2$ *is not a geodesic since* a_1 *and* a_2 *are not adjacent; but* $a_1 \to b \to c_1$ *is a geodesic of length 2). Every shortest path between distinct non-bottleneck events passes through* b*, since there are no within-layer edges.*

Betweenness: $\beta(b) \gg \beta(a_i) = \beta(c_j)$*, so* $K_{\text{geo}}(b) \gg 0$.

Free-fall analogue

All geodesics between distinct layers pass through the bottleneck b*. This means that any "motion" through the event structure from Layer 0 to Layer 2 is drawn through* b *— not by an imposed force, but by the geometry of accessibility. The bottleneck acts as a gravitational attractor in the event graph.*

Conclusion. *The three-layer bottleneck model is curved: all curvature indicators are nonzero, and geodesics concentrate through the bottleneck. This demonstrates that non-uniform event accessibility produces detectable geometric distortion.*

9 Comparison of Curvature Candidates

Table 1: Curvature indicators on the two toy models.

Indicator	Flat model ($K_{5,5}$)	Bottleneck (b)	Outer events (a_i, c_j)
K_{grow}	0	-2.24	$+0.28$
K_{branch}	0	6.2	1.0
K_{geo}	0	$\gg 0$	≈ 0

All three indicators vanish in the flat model and are nonzero in the curved model. The bottleneck event b has the strongest curvature signal in all three measures. The outer events have moderate positive volume-growth curvature (their neighborhoods are smaller than average) but low geodesic concentration (paths don't preferentially pass through them).

10 Relation to Known Curvature Concepts

Table 2: Structural parallels with known curvature frameworks.

Concept	VFD (this paper)	Comparison
Flatness	Uniform neighborhood profiles	Local isometry (Riemannian)
Volume curvature	Ball-size deviation	Bishop–Gromov comparison
Branching curvature	Degree deviation	Ollivier Ricci curvature
Geodesic curvature	Betweenness concentration	Geodesic focusing (Raychaudhuri)
Free fall	Geodesic concentration	Geodesic principle in GR

Remark 17. *These are structural parallels, not claimed equivalences. The VFD curvature indicators are discrete, graph-theoretic quantities; their continuous limits (if they exist) are not computed here. Whether they converge to smooth Ricci or scalar curvature in an appropriate refinement limit is an open question of significant interest.*

11 Discussion and Limitations

11.1 What is established

- Flatness defined as local neighborhood-profile uniformity (Definition 3).
- Three curvature indicators constructed: volume-growth, branching, geodesic concentration (Definitions 5, 8, 11).
- Flat toy model: all indicators vanish (Example 15).
- Curved toy model: all indicators nonzero, bottleneck concentrates geodesics (Example 16).
- Free-fall analogue: geodesics drawn through high-accessibility regions (Section 6).

11.2 What is not established

- No Einstein equations. The relationship between curvature and matter/energy content is not derived.
- No curvature tensor. The indicators are scalar quantities, not tensorial.
- No continuous limit. Whether the discrete curvature converges to smooth Riemann/Ricci curvature is open.

- No dynamics. How curvature evolves or responds to event-structure changes is not addressed.
- The curvature indicators are defined relative to a reference profile; the choice of reference may affect the values.
- The toy models are illustrative; the full 600-cell event structure is not analysed.

11.3 Programme position

- Paper XXIII: event ordering (derived).
- Paper XXIV: observer kinematics (derived).
- Paper XXV: separation geometry (derived).
- Paper XXVI (this paper): curvature precursor (constructed).
- Future: dynamics / field equations (how matter sources curvature).

12 Conclusion

The event-order framework of Papers XXIII–XXV naturally supports a curvature precursor. Flatness is defined as local uniformity of the transition-graph neighborhood structure; curvature is defined as deviation from that uniformity. Three indicators — volume-growth distortion, branching asymmetry, and geodesic concentration — provide complementary measures of local geometric distortion.

The flat toy model (symmetric $K_{5,5}$) has vanishing curvature. The curved toy model (asymmetric bottleneck) has nonzero curvature, with geodesics concentrating through the bottleneck in a structural analogue of gravitational free fall: geodesic propagation through the event structure follows the geometry of accessibility without any imposed force.

The GR programme now carries: temporal ordering, observer-dependent kinematics, separation geometry, and curvature. What remains is the connection between event-structure distortion and matter/energy content — the route to dynamical field equations.

References

Lee Smart. Metric emergence from event-order geometry. 2026. Paper XXV. Preprint, Institute of Vibrational Field Dynamics.